

DeriData API Integration in CRM

Bond Autofill — Two-Phase Calculator Flow (Management Brief)

Audience: Product / Ops / Leadership

Scope: CRM Bond Auto-Update Autofill only

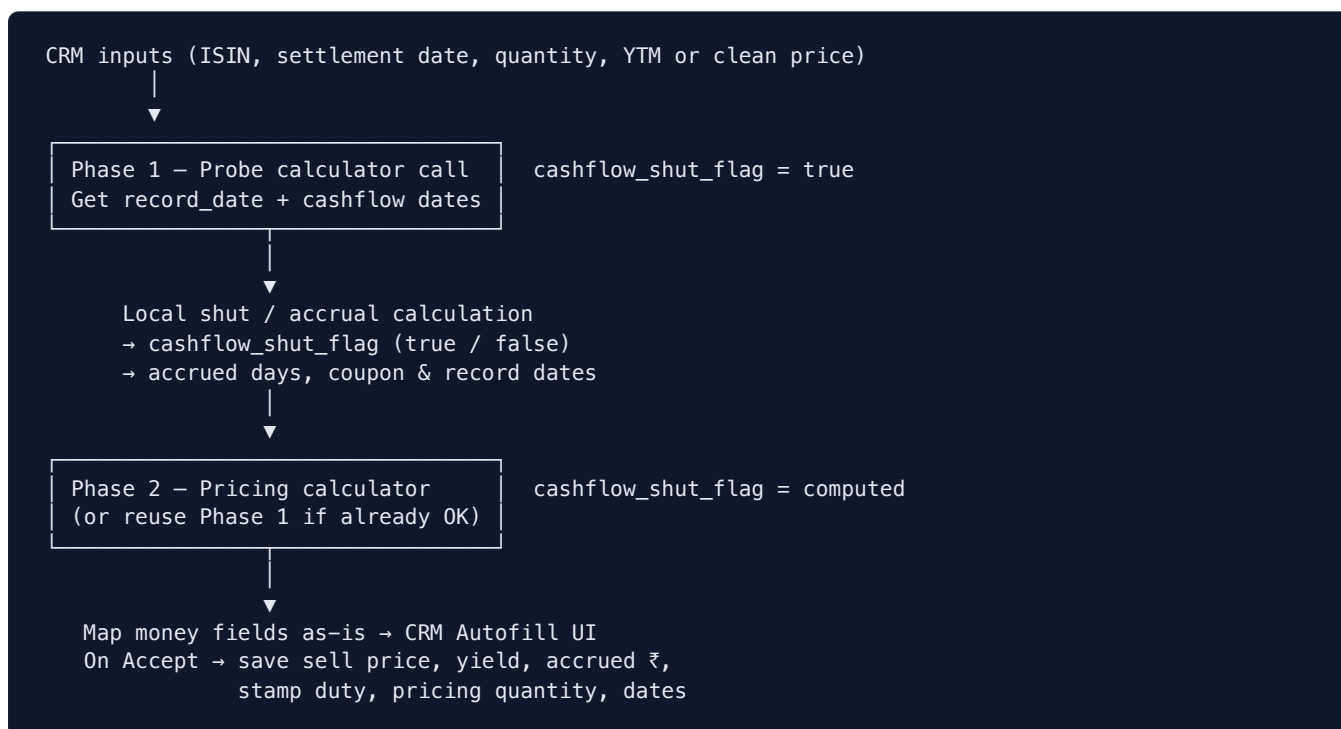
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1. Purpose

DeriData is the external bond calculator used by MeraDhan CRM to price bonds during **autofill**. Ops reviews the suggested values and Accepts them onto the bond master record. Customer place-order and draft PDFs then reuse those **saved CRM values** (scaled by quantity) — they do **not** call DeriData live.

One-line summary: We call DeriData once to learn dates and decide shut period, then call again with the correct shut flag so price, accrued interest, and settlement amounts are consistent.

2. End-to-End Flow



3. Phase 1 — Probe (Dates & Shut Decision)

API CALL 1 — SAME DERIDATA CALCULATOR ENDPOINT

Purpose: obtain calendar data needed to decide whether settlement is in the shut (ex-interest) window.

Keys sent to DeriData

Key	Source / value	Role
isin	Bond ISIN	Instrument identity
value_date	CRM settlement date	Pricing / settlement date
amount	Face value × quantity (crores)	Notional size
yield_to_price	true	YTM → price mode for probe
ytm	CRM yield (or probe default 10.5)	Required by calculator
cashflow_shut_flag	true (always for probe)	Forces cashflow/date payload
type_field	cashflow	Request cashflow schedule

Keys read from DeriData (Phase 1)

DeriData field	Used for
record_date	Shut-period start (record date)
cashflows[].cash_flow_dates	Last coupon & next coupon dates

4. Manipulation — Shut Period & Accrual Days

Between the two API calls, MeraDhan computes shut status locally from DeriData dates. This is the **only business key we override and send back** into Phase 2.

Manipulated key: `cashflow_shut_flag`

Set to our computed `isUnderShutPeriod` (not DeriData's echo of the probe input).

Shut rule

Shut is **ON** when:

`RECORD_DATE ≤ SETTLEMENT_DATE < NEXT_COUPON_DATE`

Settlement on the coupon date → shut ON, accrual days = 0.

Output key	How it is calculated
<code>isUnderShutPeriod</code> / <code>cashflowShutFlag</code>	From shut rule above
<code>accruedDays</code>	Shut: negative days from settlement → next coupon Normal: positive days from last coupon → settlement
<code>recordDate</code> , <code>lastCouponDate</code> , <code>nextCouponDate</code>	Parsed from DeriData <code>record_date</code> + cashflow dates
<code>recordDays</code>	Calendar days between record date and next coupon
<code>periodStatus</code>	"Shut Period" or "Normal"

5. Phase 2 — Pricing (Money Amounts)

API CALL 2 — SAME ENDPOINT, CORRECTED SHUT FLAG

Purpose: obtain final clean price, accrued interest, consideration, stamp duty, and settlement amount under the **correct** shut flag.

YTM mode

- `yield_to_price`: true
- `ym` = CRM pricing yield
- `cashflow_shut_flag` = **computed**

Clean-price mode

- `yield_to_price`: false
- `clean_price` = CRM clean price
- `cashflow_shut_flag` = **computed**

Reuse optimisation: If Phase 1 already used `cashflow_shut_flag` = true and our computed shut is also true (YTM mode), Phase 1's response is reused — no second network call (avoids timeouts).

Money keys mapped from DeriData (as-is)

DeriData field	CRM / saved field	Meaning
<code>summary.clean_price</code>	Sell / clean price	Clean price (% of par)
<code>summary.xirr</code>	Yield (clean-price mode)	YTM %
<code>summary.principal</code>	Principal amount	Principal ₹
<code>summary.accrued_int_top</code>	Accrued % of par	Accrued interest %
<code>summary.accrued_int_bottom</code>	<code>bonds.accruedInterest</code>	Accrued interest ₹ (saved on Accept)
<code>summary.total_consideration</code>	Total consideration	Principal + accrued
<code>summary.stamp_duty</code>	Stamp duty	Stamp duty ₹ (or derived if missing)
<code>summary.settlement_amount</code>	Settlement amount	Consideration + stamp duty

After Phase 2, principal / accrued / consideration are **not** recalculated in-house. DeriData amounts are preferred and shown in CRM autofill.

6. What Happens After Accept

- 1. Ops reviews autofill suggestions in CRM and Accepts.
- 2. Bond master is updated with sell price, yield, accrued interest (₹), stamp duty, pricing quantity, and related dates.
- 3. Customer place-order and draft order PDFs scale those **saved** values by order quantity. No live DeriData call on the customer path.

7. Why Two Calls?

Risk if we skip Phase 1	Impact
Wrong <code>cashflow_shut_flag</code> sent to pricing	Incorrect accrued interest (sign/amount) and settlement totals
Rely only on local coupon schedule	Can disagree with DeriData record date → shut mismatch
Always double-call without reuse	Slow CRM autofill / proxy timeouts under load

8. Boundary Summary

Surface	DeriData?	Pricing source
CRM Bond Autofill	Yes (two-phase)	Live DeriData + local shut calc
Place-order UI	No	Saved CRM bond values × quantity
Draft / order PDF	No	Saved CRM / order snapshot × quantity